Daily Drilling Report
SPD19B-03
8/28/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
32.0	12 1/4"	177.00	72.83	31.70	94.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	225,147.73	22,401.00	7,432,003.79		27.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,437.00	2,437.00		0.00		

Operations at Report Time
RIH directional 12 1/4" BHA#9 to 2150m

Operations Summary
Cont' hard B/R f/ 1880m to 1620m. Normal B/R f/1620m to 1490m. F/C, OK. POOH BHA#8 on elevator to surface. P/U, M/U & RIH BHA#9 to 231m.

Operations next 24hrs
Cont' drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Behnam Jahangiri/Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/28/2014	10/18/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	11:30	11....	12 1/4"	D	06	P	Hard B/R f/ 1880m to 1620m w/ 800-850 gpm/ 1500-1700 psi/ 50-60 rpm/ 12-34Kft-lbs. Normal B/R f/1620m to 1490m w/700 gpm/ 50-45 rpm/ 8-20Kft-lbs. Pump 30 bbl Lo-Vis + 30 bbl weighted pill and circulate out while B/R. Observe abundant amount of fine drill cuttings on shakers.
11:30	18:30	7.00	12 1/4"	D	04	P	F/C, Ok. Cont' POOH directional BHA#8 f/ 1490m to surface on elevator. L/D 7" Jar, Drain the 9 5/8" mud motor.B/O & L/D the 12 1/4" PDC bit (2-4-BT-N/S-X-I-LM-PR), mud motor, 12" string stab., 8 1/4" DC, NMDC. Retrieve MWD probe. R/B the rest of BHA. ** Motor bearing gap: 4mm ** 12 1/8" Sleeve stabilizer: 1/16" under gauge ** 12" String stabilizer: In gauge ** Prepared 770 bbl 12.2 ppg CaCo3 Mud ** 500 bbls KCL-Polymer 12.4 ppg mud sent to Sahar1 on Swisco Supplier.
18:30	21:30	3.00	12 1/4"	D	04	P	P/U and M/U directional BHA#9 consisting of following items. RIH to 28m. * New 12 1/4" PDC Bit (VM616PGUX-V2, IADC: M422, SN: 6011971, Nozzles: 9 x 12/32") * New 9 5/8" Mud Motor (AKO: 1 deg. , Sleeve stabilizer: 12 1/8") * New 8" Pony NMDC * New 11 7/8" String Stabilizer (COR-3103-AD) * 8" NMDC * 8" TM - Sub with Gamma and MWD probes inside
21:30	22:00	0.50	12 1/4"	D	04	P	Connect TDS and perform Motor & MWD shallow hole test w/ 800 gpm/ 1000 psi , OK.
22:00	00:00	2.00	12 1/4"	D	04	P	Cont' RIH rest of BHA items consisting of: 1 jt 8" NMDC + 8 1/2" Float sub with float valve inside + 1 jt 8 1/4" DC + X/O + 16 jts 5 7/8" HWDP + 7" Drilling Jar + 3 jts 5 7/8' HWDP to 231m.
00:00	00:00		12 1/4"				Aug 29 , 2014 00:00 - 06:00 Con't RIH directional BHA#9 on 5 7/8" DP f/ 231m to 2150m in good hole condition.

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,437.00	KCl-Polymer Mud	11.20	48	18.0	17.000	4.3
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.2	0.100	0.600	0.600			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	5	2.0	82.0	16.0	46,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
4.000	6.000	9.000	2.4/97.6		1035.0	1712.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
4956.0	78.0	0.0			6.654657	0.511

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.00	6.	135	12.0	998.0	34.5

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-HTN	Chopper	8/28/2014 11:16	8/28/2014 11:23	Pax in: 6 , Pax out: 4
Fadak	Crew Boat	8/28/2014 19:20	8/28/2014 19:55	Disembarked 01 PAX & Embarked 01 PAX for Kish
Sea Command	Supply Boat	8/27/2014 20:50		Delivered Cargo , Tranfrd FW-150 MT to Swissco Supplier. W.O.I.
Marina-1	Supply Boat	8/16/2014 15:15		At Sahar-2
Swisco Supplier	Supply Boat	8/27/2014 22:10	8/28/2014 22:30	Received FW-150 MT from Sea Command. Received 500 BBLs of Liquid Mud f/DD6. Departed for Sahar-1

Mud Additives

Description	Consumed	Cum Rec	On Loc
CaCo3-F BB	87.0	89.0	2.0
Duo-Tec	21.0	185.0	131.0
KCL	6.0	52.0	32.0
Starch	20.0	680.0	194.0
Pac-LV	20.0	262.0	15.0
Zinc Carbonate		119.0	119.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Drill Water	3.0	0.0	330.0
Potable Water	20.0	18.0	108.0
Diesel Fuel	7.0	0.0	183.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	207.0
Bentonite BB	0.0	0.0	0.0
Cement G	0.0	0.0	132.0
Bentonite Bulk	0.0	0.0	8.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
8	12 1/4	BEST	M1385	228378	PDC	M432	12/12/12/12/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.95	94.80	748,181.40	1,495.00	2,437.00	942.00	9.9	2-4-BT-N/S-X-I-LM-PR
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	880	1,050.0	1,920.0	9	37	97	179
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
142.30	58.89	62.12	37.88	382	342	243	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,437.00	2,437.00							

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.250	27.83	226.57
7" Drilling Jar (COR-3022-AD)	7	3.000	5.85	198.74
16 jts 5 7/8" HWDP	5 7/8	4.250	149.23	192.89
X/O	7	2.810	0.79	43.66
1 jt 8 1/4" DC	8 1/4	2.810	9.46	42.87
8 1/2" Float Sub	8 1/2		0.79	33.41
8" NMDC	8	3.500	9.44	32.62
8" TM - Sub	8		2.03	23.18
8" NMDC	8	3.500	9.46	21.15
12" S. Stabilizer (COR3059AD)	11 7/8	3.125	2.19	11.69
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.14	9.50
12 1/4" PDC bit	12 1/4		0.36	0.36

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
9	12 1/4	Varel	VM616PGUX-V2	6011971	PDC	___	12/12/12/12/12/12/12/12
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.99	0.00						-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
0.11							

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Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.83	231.39
7" Drilling Jar (COR-3022-AD)	7	2.938	5.85	203.56
16 jts 5 7/8" HWDP	5 7/8	4.000	149.23	197.71
X/O	7	2.813	0.79	48.48
1 jt 8 1/4" DC	8 1/4	2.810	9.46	47.69
8 1/2" Float Sub	8 1/2		0.79	38.23
8" NMDC	8	3.500	9.44	37.44
8" TM - Sub	8		2.03	28.00
8" NMDC	8	3.500	9.46	25.97
11 7/8" S. Stabilizer (COR3103AD)	11 7/8	2.813	2.16	16.51
8" PNMDC	8 1/16	3.500	4.03	14.35
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.95	10.32
12 1/4" PDC bit	12 1/4		0.37	0.37

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
L.Sur.Sh.		2,161.00	2,055.29
Lithiotis Bed.		2,218.50	2,108.29
Neyriz		2,240.00	2,128.14
Up Dashtak		2,266.00	2,152.17
Evaporite "B"		2,338.00	2,218.78

General Remarks

Comment

>>> Requirements: (Remaining items of MI MR#13) + (60 MT Cement).

>>> Sent:
** Swisco Supplier to Sahar 1: (500 bbls of 12.4 ppg KCL-Polymer mud).

>>> Received: Nil.